

Top IoT development hardware and platforms to look out for

Your next IoT project need not be daunting, if you know which platform to pick

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IoT has been a buzzword for close to a decade now but for the common folk, we've only been introduced to the consumer end of IoT devices for the past 2-3 years. We're talking about the convenient Amazon Echo, Google Home and Apple HomeKit devices which have only gained popularity in recent times. It's not that these devices popped up just three years ago, they were available a little before that but with barely any proper applications and a scarce ecosystem to support the devices, it didn't make sense to get any of them. Only early adopters and folks with money to burn would have bought them.

Things have changed massively over these couple of years. The SDKs have matured and more companies have joined the fray. Also, the industry has picked their favourites, so we now have a few hardware dev boards that are favoured more than others and the same goes for SDKs. And it goes without saying that the folks who were first on the scene have gained the largest user base and the ones on this list tend to be more of those. The recent entrants onto the

scene in this list are those who've made quite the impact. This could be because of a unique feature set or simply because they did things better. The following list elaborates upon the top hardware and software IoT tools that you should consider before checking out anything else.

ARDUINO

Arduino is inarguably the most successful IoT platform in recent times. Started in 2005 by folks over at the Interaction Design Institute Ivrea, Italy, Arduino has since then matured into a massive scene. The board not only offered modular components in the form of sensors and feature extensions but because it was open-source, a lot more projects got tacked on or used Arduino as the base.



In part, the Arduino IDE also played a very important part since it made programming immensely simple for beginners to understand. Getting a microcontroller and using a USB interface to upload sketches (Arduino programs) in a matter of seconds without having to worry about configuring drivers and building the breakout board made things very simple to comprehend. Moreover, a simple format for HATs

and Shields meant that an insane number of attachments could be paired with a single dev board and even multiple could be daisy chained to rapidly prototype projects. You could literally move from idea to functional prototype in a matter of hours. And if you needed more horsepower, you didn't necessarily have to redesign from scratch. All you had to do was remove the base Arduino board and switch it out for one with a more powerful microcontroller.

Arduino has several new designs which are more in tune with the kind of hardware that smart IoT devices need. The Arduino Nano 33 IoT and the Arduino MKR family come with on-board connectivity modules with interfaces such as Wi-Fi and GSM and they're very compact. This removes the need for developers to incorporate a separate wireless connectivity shield.

RASPBERRY PI

While Arduino started off with simple microcontrollers aimed at projects that were power efficient and didn't need a lot of compute power, Raspberry Pi went the other way. A Raspberry Pi board was basically a fully fledged desktop computer fitted onto a PCB the size of a credit card. A Raspberry Pi board had network connectivity, Bluetooth, a couple of USB ports, audio port and even an HDMI port for display. A lot of low-cost computers built on Raspberry Pi boards have enabled children in developing countries to get access to their first ever computers.

Like the Arduino, the folks behind Raspberry Pi were also focused on